

Harpreet Singh

Embedded Hardware Designer

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Summary

Embedded hardware designer with 13+ years across schematic design, PCB design, board bring-up, mixed-signal acquisition, SMPS/power electronics, embedded interfaces, sensor/actuator hardware, and hardware/software validation. Built telecom, media, LiFi, test-equipment, EEG/fNIRS, STM32, Raspberry Pi, and FPGA-adjacent systems using practical bench-debug workflows with oscilloscopes, logic analyzers, JTAG/SWD, SPI, I2C, UART, CAN, Ethernet, and power-management validation.

Core Skills

- Schematic design, PCB design, multilayer boards, component selection, BOM review, board bring-up, rework/debug planning
- SMPS design, isolated DC/DC sections, power sequencing, power budgeting, regulators, watchdog PWM, grounding and signal integrity awareness
- Mixed-signal acquisition, ADS1299, ADS8688, DAC8565, sensor interfaces, analog sampling, MUX/TDM control, Raspberry Pi hardware validation
- ARM Cortex, STM32, AVR, PIC, ESP32, MicroPython boards, JTAG, SWD, oscilloscopes, logic analyzers, hardware test scripts
- SPI, I2C, UART, CAN, RS-485, Modbus, USB, Ethernet/LWIP, GPIO, interrupts, embedded C/Python hardware test automation

Professional Experience

Research Assistant, Core-Hub Neuroengineering Solutions | *University of Lethbridge* | Apr 2025 - Present

- Design and validate embedded neuroengineering hardware for EEG/fNIRS acquisition, mixed-signal control, Raspberry Pi SPI/GPIO interfaces, sensor inputs, output control, and bench testing.
- Developed an integrated 8-channel EEG + 8-channel fNIRS acquisition platform using ADS1299, ADS8688, DAC8565, watchdog PWM, TDM MUX sequencing, shared SPI0, and GPIO control lines.
- Documented hardware wiring, chip-select mapping, DRDY/RESET/START signals, MUX control lines, CSV test outputs, and safe bring-up checks for live acquisition experiments.

Research Assistant | *University of Regina* | Jan 2021 - Apr 2024

- Researched uncertain-reasoning intrusion detection for DoS/DDoS detection using Bayesian Networks, Markov Networks, Zeek, Wireshark, feature engineering, and Python ML workflows.
- Published SECRIPT 2024 work on uncertain-reasoning IDS methods and presented findings to an international cybersecurity research audience.
- Built ML prototypes including a Bayesian liver-disorder diagnostic model with 85% accuracy and Punjabi BERT/NLP classifiers for 100K+ news articles with 92% classification accuracy.

Independent Embedded Systems Developer | *Self-Employed* | Jun 2024 - Mar 2025

- Designed a custom STM32F405RGT6 MicroPython development board, including multilayer PCB design, firmware flashing path, sensor/actuator interfaces, power management, and hardware/software debugging.

Senior Software Engineer | *Planetcast Media Services Ltd.* | Dec 2018 - Nov 2020

- Led a 15-member team delivering embedded hardware and firmware for broadcast/media systems, including a network-controlled 10-channel video switcher and Ethernet temperature acquisition platform.
- Designed schematics and PCBs, selected components, coordinated cabinet/mechanical integration, and validated control/alarm interfaces with Ethernet, HTTP/LWIP, and embedded C firmware.

Embedded Software Engineer / R&D Engineer | *Spark Eighteen Pvt. Ltd.; Exicom Tele-systems Ltd.* | Feb 2016 - Dec 2018

- Designed and brought up STM32/AVR embedded hardware for LiFi control, telecom power plants, DC interface cards, and automatic test equipment.
- Designed 15-20 W isolated SMPS sections with 78-81% measured efficiency and debugged boards across CAN, SPI Flash, I2C EEPROM, Ethernet/LWIP, GUI test automation, and industrial power-monitoring interfaces.

Selected Projects

BCT_8EEG_8FNIRS

- Designed and documented a mixed-signal EEG/fNIRS hardware stack using ADS1299 8-channel EEG acquisition, ADS8688 8-channel analog acquisition, DAC8565 output control, shared SPI0, GPIO-controlled MUX/TDM sequencing, watchdog PWM, and Raspberry Pi bench-test software.

PyBoard / STM32 MicroPython Board

- Designed a custom STM32F405RGT6 development board with multilayer PCB design, MicroPython firmware support, sensor/actuator interfaces, power-management debugging, firmware flashing, and board bring-up planning.

Industrial Embedded Hardware

- Delivered hardware for telecom, LiFi, media, Ethernet switching, temperature acquisition, DC power monitoring, automatic test equipment, and isolated 15-20 W SMPS sections with 78-81% measured efficiency.

Publication

- Harpreet Singh et al., "An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection," SECRIPT 2024, ISBN 978-989-758-709-2, ISSN 2184-7711, pp. 771-776.
- Published thesis: Singh H. A robust intrusion detection system utilizing uncertain reasoning techniques in artificial intelligence. The University of Regina (Canada); 2024.

Education

- PhD in Computational Behavioural Neuroscience, University of Lethbridge - In progress
- M.Sc. Computer Science, University of Regina - 2024
- Bachelor's equivalent in Electronics & Telecommunication Engineering, AMIETE/IETE - 2012